STUDENT PROJECTS LABORATORY - LOCKOUT / TAGOUT [LOTO] PROCEDURE

ENERGY ISOLATION SPECIFICATION | ASSET: CNC-02 | TAG: 01191439

Haas VF-2SS CNC Vertical Machining Center

Authorized LOTO Personnel: Sr. Instructional Lab Supervisor (Max Wright) & Student Research Assistants (SRAs)

Notification Channel: Max Wright - Teams / Text Message | Zone: Zone 2: CNC Machining (CNC)

STEP 1 - NORMAL MACHINE SHUTDOWN PROCEDURE

Depress E-Stop and Red Power Off button (Front right control panel).

STEP 2 - HAZARDOUS ENERGY SOURCES, MAGNITUDES & ISOLATION POINTS

Source #1: Electrical (240V AC 3-Phase)

Isolation Location: Main circuit breaker on rear cabinet / wall disconnect.

Lockout Procedure: Switch Main circuit breaker to OFF position with individualized lock & tag; WAIT 5 MINUTES for vector drive capacitors to dissipate and verify vector drive voltage LED is completely OFF.

Source #2: Pneumatic (80 PSI Compressed Air)

Isolation Location: Behind machine on wall air drop.

Lockout Procedure: Turn red handle counterclockwise to OFF, bleed bottom filter valve, and verify pressure gauge reads 0 PSI.

STEP 3 - STORED / RESIDUAL ENERGY DISSIPATION

Mandatory 5-minute wait for vector drive capacitor discharge (verify LED off) + pneumatic line bleed to 0 PSI.

STEP 4 - ZERO-ENERGY VERIFICATION ('TRY' STEP)

Press [POWER ON] on pendant and verify with non-contact AC voltage detector at input terminals + confirm air gauge reads 0 PSI.